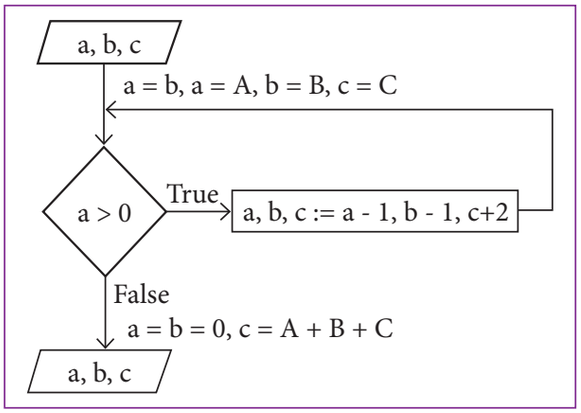
In each meeting, ***a*** and ***b*** each decreases by ***1***, and ***c*** increases by ***2***. The solution can be expressed as an iterative algorithm.

```
monochromatize (a, b,  c)
      -- inputs : a = A, b=B, c=C, a=b
      -- outputs : a = b = 0, c = A+B+C
while a>0
       a, b, c := a-1, b-1, c+2
```

The algorithm is depicted in the flowchart as below



Now let us write this algorithm using function

```
let rec monochromatize a  b  c :=
      if a > 0 then
        a, b, c := a-1, b-1, c+2
      else
        a:=0, b:=0, c:= a + b + c
        return c
```

Points to remember:

- Algorithms are expressed using statements of a programming language
- Subroutines are small sections of code that are used to perform a particular task that can be used repeatedly
- A function is a unit of code that is often defined within a greater code structure
- A function contains a set of code that works on many kinds of inputs and produces a concrete output
- Definitions are distinct syntactic blocks
- Parameters are the variables in a function definition and arguments are the values which are passed to a function definition through the function definition.
- When you write the type annotations the parentheses are mandatory in the function definition
- An interface is a set of action that an object can do
- Interface just defines what an object can do, but won't actually do it
- Implementation carries out the instructions defined in the interface
- Pure functions are functions which will give exact result when the same arguments are passed
- The variables used inside the function may cause side effects though the functions which are not passed with any arguments. In such cases the function is called impure function

XII Std - CS EM Chapter-1.indd 7 24-01-2020 12:38:55